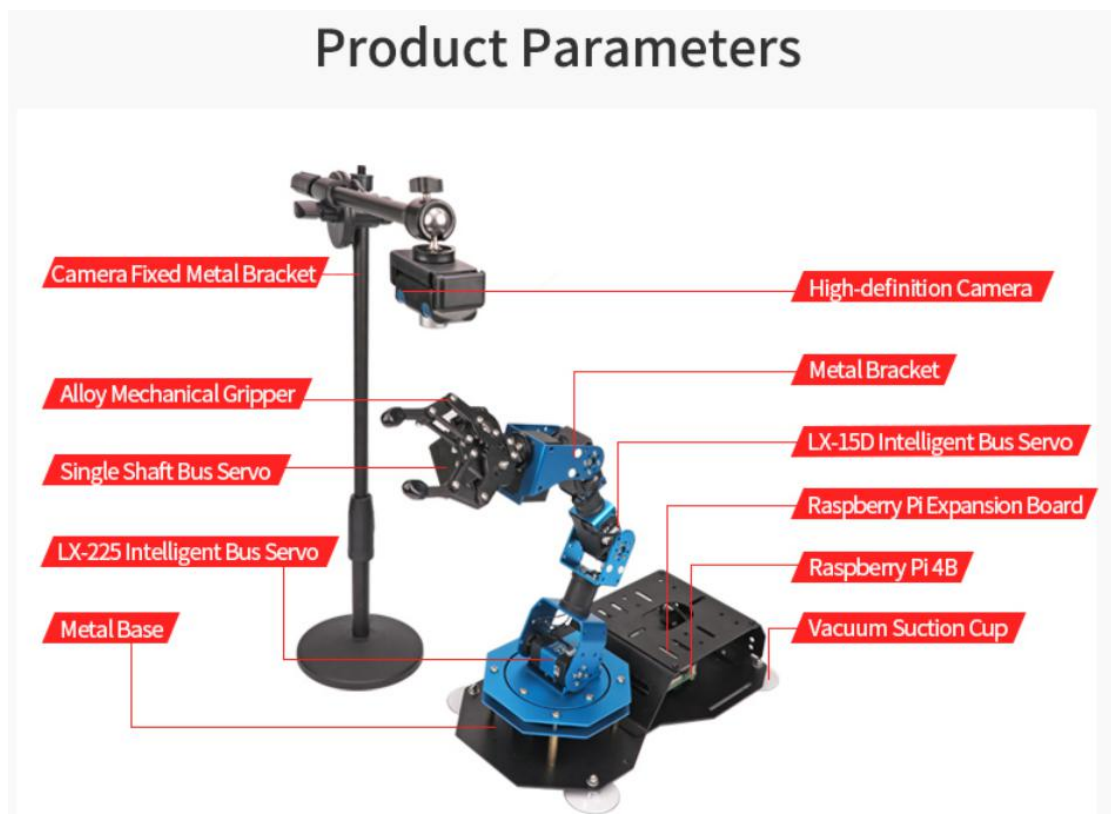


Lesson 1 ArmPi Introduction

1. Introduction

The intelligent visual robotic arm ArmPi is powered by Raspberry Pi 4B. With built-in inverse kinematics algorithm and a high-definition camera. ArmPi can sort color blocks, track and grab, and stack blocks intelligently.



For better experience, we have prepared abundant tutorials, software and open source of programming games. The system images are burned before delivery.

2. Overview of the Learning Tutorials

Step1: ArmPi Instruction

Folder “1.ArmPi Introduction” explains the structure, assembly, software and how to start the ArmPi.

Step2: Quick Start Instructions

Please go to the folder “2.Quick User Experience” to learn how to calibrate position and the APP control.

The connected method of “LAN Mode” is an optional content, users can learn as needed.

Step3: Getting Ready for AI Vision Programming Games

Learn the installation of VNC and device connection. The frequency modification method is for your reference.

The function of “Start position correction operation via command line” is the same as “Position Calibration” in the folder “2.Quick User Experience”. The difference is that the former sends command line to operate.

“Frequency Modification Method” is suitable for all the users who have no 5G network card on their computer and provide a solution to modify the frequency when the device cannot be found to turn on the hotspot. (Note: The ArmPi defaults to the 5G frequency, which has a better experience than the 2.4G frequency. Users who use desktop computers are advised to prepare a 5G frequency network card.)

Step4: AI Vision Programming Learning

Learning the command line startup method of each AI Vision games to have a brief understanding of the implementation process. In addition, users can modify the program with their ideas.

Compared with the mobile phone APP games, it can further deepen the experience of OpenCV machine vision and robotic arm inverse kinematics.

Step5: Advanced Learning

1. For the tutorials of PC Software introduction, action group programming, execute actions and integrate action groups, please go to the folder “4.Advance Program Lessons/1. ArmPi Action Programming”.

2. Raspberry Pi Controller Learning

With this chapter, you will have an idea of what is Raspberry Pi and the basic operation.

3. Raspberry Pi Expansion Board Learning

This chapter explains the working principle of bus servo, buzzer and RGB lights with the examples.

4. OpenCV Basic Lesson

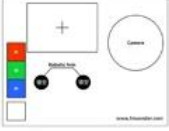
Brief introduction of machine vision OpenCV, which is useful for the future learning.

5. Inverse Kinematics Lesson

This chapter mainly helps users to learn the trajectory planning and control of ArmPi manipulator (ArmPi mainly uses the inverse kinematics part).

We provide formula algorithms of inverse kinematics, and mark the source code path of related programs. If you have any interests, please check the code comments.

3. Packing List

ArmPi Rotic Arm Packing List			
			
ArmPi robotic arm	7.5V 6A power supply adapter	Camera fixed metal bracket	Camera
			
Map	3*3cm red, green and blue blocks	16G U disk	Card reader